

Trick of squares of Numbers Consisting 9s

$$\underbrace{9999}_\text{One less 9 than original}^2 = \underbrace{999}_\text{Same number of 9 and 0} \underbrace{8000}_\text{Same number of 9 and 0} \underbrace{1}_\text{Same number of 9 and 0}$$

Stuff 8 and 1
between 0s.
For square,
 $8+2=10$
and $8+1=9$

Total number of digits in the expression = Double the total number of 9s

$$\text{Digits Sum} = 9 \times \text{Total Number of 9s}$$

Trick of cubes of Numbers Consisting 9s

$$\underbrace{9999}_\text{One less 9 than original}^3 = \underbrace{999}_\text{Same number of 9 and 0} \underbrace{7000}_\text{Same number of 9 and 0} \underbrace{2999}_\text{Same number of 9 and 0}$$

Stuff 7 and 2
between 0s
For cube,
 $7+3=10$
and $7+2=9$

Append by 1 in original number of 9

Total number of digits in the expression = Triple of total number of 9

$$\text{Digits Sum} = 9 \times \text{Total Number of 9s} \times 2$$

Trick of Squares of Numbers Consisting 1s

$$(\text{Number of 1s})^2 = 123 \dots \text{max number} \dots 123$$

$$\boxed{\quad} = \boxed{\quad}$$

Special Property of Table of 37

$$37 \times 3 = 111$$

$$37 \times 6 = 222$$

$$37 \times 9 = 333$$

$$37 \times 12 = 444$$

$$37 \times 15 = 555$$

...

$$\text{Sum of all factors} = \left(\frac{p_1^{a+1} - 1}{p_1 - 1} \right) \times \left(\frac{p_2^{b+1} - 1}{p_2 - 1} \right) \times \left(\frac{p_3^{c+1} - 1}{p_3 - 1} \right)$$

$$\text{Product of all factors} = N^{\frac{T}{2}}$$

Trick for Compound Interest

$$2 \text{ years} \longrightarrow 2 : 1$$

$$3 \text{ years} \longrightarrow 3 : 3 : 1$$

$$4 \text{ years} \longrightarrow 4 : 6 : 4 : 1$$

Time and Work Tricks

Product Sum Method
(PDS - Product divided by Sum)
(Valid only for two people)

$$A \longrightarrow x \text{ days}$$

$$B \longrightarrow y \text{ days}$$

$$A + B = \frac{\text{Product}}{\text{Sum}} \text{ days}$$

Product Difference Method
(PDD - Product divided by Difference)

$$A + B = p \text{ days}$$

$$B = q \text{ days}$$

$$A = \frac{\text{Product}}{\text{Difference}} = \frac{pq}{q-p}$$

PDS for three person

$$A \longrightarrow x \text{ days}$$

$$B \longrightarrow y \text{ days}$$

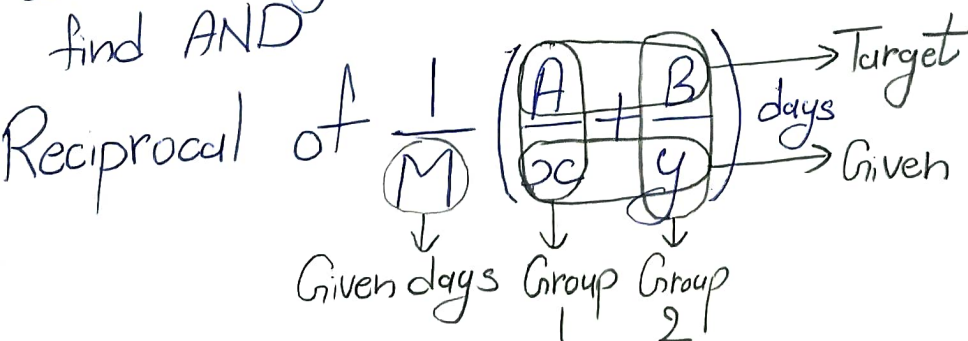
$$C \longrightarrow z \text{ days}$$

$$A + B + C$$

$$= \frac{xyz}{xy + yz + zx} \text{ days}$$

OR-AND Problems

If OR is given and we have to find AND



$$A + B \longrightarrow x \text{ days}$$

$$B + C \longrightarrow y \text{ days}$$

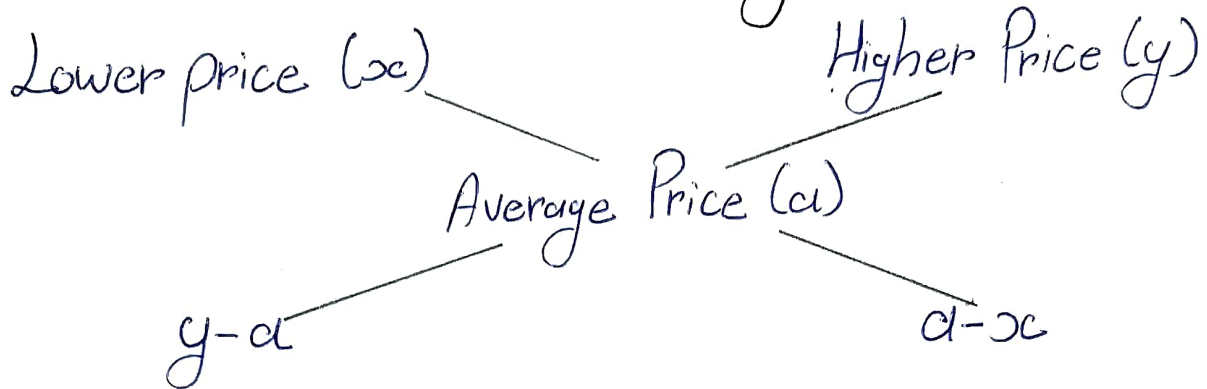
$$C + A \longrightarrow z \text{ days}$$

$$A + B + C = \frac{2xyz}{xy + yz + zx} \text{ days}$$

Work Leaving Problem

$$\frac{\text{Actual Time A worked}}{\text{Time taken if A worked alone}} + \frac{\text{Actual Time B worked}}{\text{Time taken if B worked alone}} + \dots = 1$$

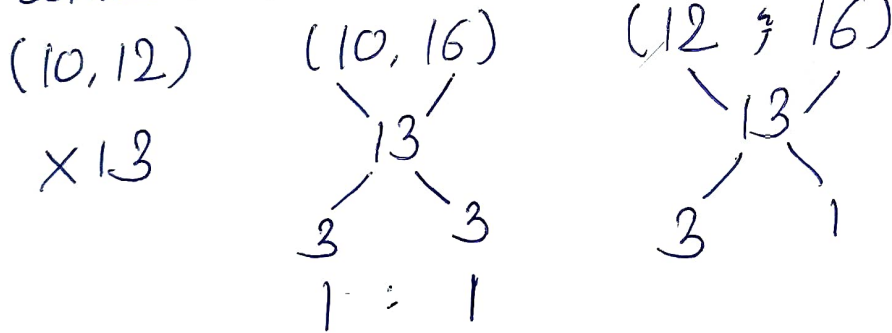
Mixture and Alligations



Trick for 3 ingredients:

We have 10, 12 and 16, we can form pairs
(10, 12), (10, 16), (12, 16)

We can reject pair (10, 12) because average 13 does not come in it



$$10 \Rightarrow 1, \quad 12 \Rightarrow 3, \quad 16 \Rightarrow 1+1=2$$

Ratio is 1:3:2